International Journal of Sports Physiology and Performance, 2019, 14, 1447-1454
https://doi.org/10.1123/ijspp.2018-0874
© 2019 Human Kinetics, Inc.

Human Kinetics
INVITED COMMENTARY

Does Mathematical Coupling Matter to the Acute to Chronic Workload Ratio? A Case Study From Elite Sport

Joseph O.C. Coyne, Sophia Nimphius, Robert U. Newton, and G. Gregory Haff

Purpose: Criticisms of the acute to chronic workload ratio (ACWR) have been that the mathematical coupling inherent in the traditional calculation of the ACWR results in a spurious correlation. The purposes of this commentary are (1) to examine how mathematical coupling causes spurious correlations and (2) to use a case study from actual monitoring data to determine how mathematical coupling affects the ACWR. ***Methods:*** Training and competition workload (TL) data were obtained from international-level open-skill (basketball) and closed-skill (weightlifting) athletes before their respective qualifying tournaments for the 2016 Olympic Games. Correlations between acute TL, chronic TL, and the ACWR as coupled/uncoupled variations were examined. These variables were also compared using both rolling averages and exponentially weighted moving averages to account for any potential benefits of one calculation method over another. ***Results:*** Although there were some significant differences between coupled and uncoupled chronic TL and ACWR data, the effect sizes of these differences were almost all trivial ($g = 0.04$–0.21). Correlations ranged from $r = .55$ to .76, .17 to .53, and .88 to .99 for acute to chronic TL, acute to uncoupled chronic TL, and ACWR to uncoupled ACWR, respectively. ***Conclusions:*** There may be low risk of mathematical coupling causing spurious correlations in the TL–injury-risk relationship. Varying levels of correlation seem to exist naturally between acute and chronic TL variables regardless of coupling. The trivial to small effect sizes and large to nearly perfect correlations between coupled and uncoupled AWCRs also imply that mathematical coupling may have little effect on either calculation method, if practitioners choose to apply the ACWR for TL monitoring purposes.

Keywords: training load, spurious correlations, monitoring, statistics

Monitoring training and competition workload (TL) has become increasingly popular in sport research focusing on decreasing the risk of injury.[1,2] One variable that has received increasing interest is the acute to chronic workload ratio (ACWR). The ACWR was originally calculated as the 7-day acute rolling average (ATL) divided by the 28-day chronic rolling average (CTL), although several studies have assessed the ACWR using weekly data.[1,3] The use of the ACWR is contentious, with 1 criticism of the ACWR suggesting that the mathematical coupling in the calculation of the ACWR (ie, the numerator ATL is contained in the denominator CTL) causes a spurious correlation.[3] A spurious correlation occurs when 2 variables are incorrectly inferred to be related to one another or the relationship is incorrectly understated or overstated.[4] A potential cause of this type of error is mathematical coupling. There are many different types of mathematical coupling (described previously),[4] which can cause errors. In regard to the ACWR, Lolli et al[3] demonstrated differences between a coupled ATL–CTL correlation ($r = .52$) and an uncoupled ATL–CTL correlation ($r = .01$) using randomly generated data. The coupled CTL was calculated as a 28-day average that included the most recent 7-day ATL (ie, ATL included in a 28-d CTL). Meanwhile, the uncoupled CTL was calculated as a 21-day average that excluded the most recent 7-day ATL (ie, ATL excluded from a 21-d CTL). From the differences in previously mentioned correlations, it was suggested that ATL and CTL should be uncoupled if practitioners wish to use the ACWR to examine the relationship between TL and injury.[3] The authors know of only 3 studies, all in open-skill sports, which have employed an uncoupled ACWR to examine this relationship to injury risk.[5–7] The first study on youth professional soccer players found no association between ACWR and injury likelihood.[5] By contrast, in the second and third studies, different ACWRs were associated with injury likelihood in senior professional basketball players and English Premier League soccer players.[6,7]

These studies used different injury classifications which along with the different maturation levels of the athletes may explain the differences in findings.[5,8] It also remains unclear how using a coupled ACWR would have influenced the results of both these studies. Furthermore, it is worth considering that an uncoupled ACWR does not compare the same point in time; that is, there is a lag period that disregards the most recent work an athlete completes as contributing to CTL. This may have conceptual issues considering prior recommendations for statistical models to align with theoretical models.[9] In this particular case, the ACWR is derived from the Bannister fitness-fatigue model,[10] which does not have this lag period. The issue with the lag in CTL is even more pronounced if exponentially weighted moving averages (EWMA) are preferred over rolling averages for theoretical or sensitivity reasons,[11,12] as they weight the most recent work an athlete completes.

Furthermore, a measure should not be automatically discarded, if its components are mathematically coupled.[4,13] As long as the nature of the mathematical coupling in the ACWR is accounted for and understood, it may not be problematic.[4,13] In addition, the differences between coupled and uncoupled ATL–CTL correlations may not have the ability to determine if the relationship between the ACWR and injury risk is spurious. The authors agree that a correlation between ATL and CTL when mathematically coupled may be spurious if we *solely* examined the relationship between CTL (as an explanatory variable) and ATL (as the

The authors are with the Centre for Exercise and Sports Science, School of Medical and Health Sciences, Edith Cowan University, Joondalup, WA, Australia. Coyne (coach@josephcoyne.com) is corresponding author.

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

response variable) or vice versa. However, researchers are not concerned with the ATL–CTL relationship in isolation, and the actual relationship of interest's explanatory and response variables are the ACWR and injury risk, respectively. In this relationship, there is no inherent mathematical coupling, as measures of injury risk are not overtly or even indirectly obtained from TL measures (ie, ATL, CTL, or ACWR). Therefore, it is plausible a spurious correlation between the ACWR and injury risk cannot exist.

To examine this issue further with data, a more appropriate investigation may be required to compare correlations between coupled and uncoupled ACWRs rather than coupled and uncoupled ATL–CTL correlations. It should also be noted that randomly generated data used in some of the previous research might not be the best representation of TL data when considering the effect that injuries have on TL and the repeated training cycles seen in many sports. The pattern of repeated training cycles within certain training phases (eg, general preparation, specific preparation, competition) is especially common in closed-skill sports that may only compete 4 to 6 times per year (eg, weight lifting, athletics, swimming). Meanwhile, open-skill sports like basketball, football, and rugby union also possess similar training cycles within certain training phases (eg, offseason, preseason, in-season); especially, when there are consistent time frames between competitions. As a generalization, TL measures will generally gradually trend to a peak across a number of weeks in the preparation/preseason period and decline or remain reasonably constant during a competition period. Due to these repeated training cycles in both open- and closed-skill sports, a correlation may naturally exist between ATL and CTL regardless of coupling.

However, more recently Lolli et al[14] have provided open-skill sport data (senior professional football) that displayed a trivial ATL–uncoupled CTL correlation ($r = -.04$), although preseason data were excluded from analysis which may have impacted the results. It is notable that Lolli et al[14] used total running distance (an external TL measure) as the TL variable in their analysis. Examining coupled/uncoupled TL correlations with internal and/or external TL measures (which have been described previously[15]) may give different results. These results may also be influenced by whether TL models analyze single TL variables versus training impulse (TRIMP). The TRIMP is the product of TL intensity and duration/volume factors and may be a more appropriate variable for analysis considering the conceptual origin of the ACWR.[10] Finally, and as alluded to previously, injury and illness incidents should also be accounted for in any investigation, as they will affect the regularity of training cycles and in turn, the correlation between ATL and CTL.

Based on (1) the theoretical issues in determining if the ACWR has a spurious relationship with injury risk; (2) the potential differences between TL variables in models (eg, internal, external or TRIMP); and (3) the different calculation methods of ATL/CTL; more research is needed to determine the effects of coupling in the ACWR calculation. Therefore, the purpose of this commentary is to add to the knowledge base on differences between coupled/uncoupled ATL, CTL, and ACWR across internal and external TL measures using a case study of international-level athletes in both open- and closed-skill sports.

Methods

Participants

Two groups of participants were analyzed: (1) an open-skill sport group (OS) and (2) a closed-skill sport group (CS). The OS contained 17 international-level female basketball athletes (27.8 [3.6] y, 185.0 [8.9] cm, 77.7 [10.3] kg, n = 12 with over 12 mo experience on the national team). CS contained 28 elite male (n = 13) and female (n = 15) weight lifters (26.0 [3.2] y, 162.2 [11.3] cm, 72.2 [23.8] kg, previous 12-mo personal best total 96.3% [2.7%] of world record total). The data for this study were initially collected within the athletes' training environment and released de-identified from the athletes and respective national Olympic Committee. Approval for this investigation was granted by the Edith Cowan University human ethics committee (Approval #19521) and conforms to the code of ethics of the World Medical Association (Declaration of Helsinki). Five subjects (basketball, n = 2; weight lifting, n = 3) were excluded from any analysis due to poor compliance with the data collection process.

Design

This case study contains observational TL and injury/illness data that was collected using established methods.[16–18] The data were collected prior to each group's respective Olympic qualifying events for the 2016 Rio Olympic Games. No interventions or modifications of training based on the observed TL data were made during the data collection period.

Methodology

Both internal and external TL variables were analyzed as TRIMP. Internal TL TRIMP for both OS and CS was the product of sessional ratings of perceived exertion and session duration (t).[16] For OS, external TL TRIMP was the product of PlayerLoad/minute derived from Catapult accelerometers and session duration (t).[17] PlayerLoad is an arbitrary unit derived from triaxial measures of rate of change in acceleration.[18] Due to intellectual property reasons, CS (weight lifting) would not consent to the release of their external TL data, and as such, only internal TL data were analyzed. Another difference between groups was that the OS TL data included competition workloads (ie, games/tournaments prior to the qualifying events for the 2016 Rio Olympic Games). Illness or injury incidents that caused an athlete to modify technical training or seek medical attention were also collected.[19,20] If athletes were absent from training due to injury/illness, TL was recorded as zero to enable continuous calculations of TL. Similar to previous research, ATL and CTL, respectively, required at least 7 and 28 consecutive TL daily measures. The OS data set represented a typical international-level open-skill sport's schedule where multiple tournaments were contested culminating with the 2016 Olympic qualification tournament. The OS data set also included a transition week after one of the preliminary tournaments where no training was undertaken. The practice of using recovery weeks is commonplace in the periodization literature and reflects a common training practice in an open-skill sport at this level. Conversely, the weight lifting data set seemed characteristic of a closed-skill sport with uninterrupted training (besides injury/illness incidents) leading into the penultimate competition before the 2016 Olympics.

Statistical Analysis

Statistical analyses were performed using statistical software (R statistics package, https://www.r-project.org). The full R console code has been provided as Supplementary Material (available online) for reference. The ATL, CTL, uncoupled CTL, ACWR, and an uncoupled ACWR (ATL/uncoupled CTL) were calculated using previously described methods.[1,11] These variables were assessed as both rolling averages and EWMA to account for any

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

potential benefits of TL expressed as EWMA.[1,11] A novel feature of these calculations was the use of an uncoupled CTL EWMA, which was calculated using a 21-day time constant in the same manner as Lolli et al[3] (ie, distinct ATL–CTL periods). Any missing external TL data due to technical issues with the Catapult accelerometers were estimated using individual athlete linear regression with sessional ratings of perceived exertion as the predictor and PlayerLoad/minute as the response. The percentage of modified training due to injury/illness compared with total training time was also assessed across both groups.[21] After the data were assessed for normal distribution using a Shapiro–Wilk test, coupled and uncoupled variations of CTL and ACWR were compared using a Mann–Whitney *U* test to determine if there were any significant differences between the variables. The alpha level for significance was defined as $P \leq .05$. Hedge effect sizes (g) were then converted to ascertain the effect of any difference.[22] ATL–CTL and ATL–CTL uncoupled, ACWR–ACWR uncoupled correlations were also compared across internal and external TL data. The R–z transformations were further utilized to assess any differences between correlations. Magnitude of effect sizes and correlations were interpreted as per the recommendations of Hopkins et al.[23]

Results

A total of 2861 training sessions were included in this analysis (OS n = 1717, CS n = 1144). The total amount of modified training due to injury/illness was 40.0% (34.0%) in OS and 32.0% (40.0%) in CS, which was similar across both groups ($P = .51$, $g = 0.22$). Descriptive statistics of TL variables calculated using rolling averages and EWMA are presented in Tables 1 and 2, respectively. There was a significant difference between rolling average coupled and uncoupled CTL in the OS external TL measures ($P = .03$). There were also a number of significant differences between all the EWMA coupled and uncoupled CTL variables ($P < .01$). However, despite being significant, the effect sizes of these differences were all trivial with the exception of one. The rolling average data set effect sizes ranged from $g = 0$ to 0.14. Meanwhile, the EWMA data set effect sizes ranged from $g = 0$ to 0.24 with the difference between CS internal EWMA CTL and uncoupled CTL being the variable possessing a small effect.

The results of the correlational analysis for the CS and the OS are presented in Figures 1–3. There was a range of small to large correlations between ATL and CTL uncoupled correlations across rolling averages ($r = .17$–$.35$) and small to moderate correlations across EWMA ($r = .19$–$.35$). ACWR–ACWR uncoupled correlations were nearly perfect ($r = .96$–$.99$) when calculated by rolling averages and varied from large to very large with EWMA ($r = .85$–$.94$). Using r–z transformations, ATL–uncoupled CTL correlations were only significantly different in the CS group for both rolling average and EWMA ($P < .05$).

Discussion

The key findings of this case study suggest that there may be a natural correlation between ATL and CTL that exists irrespective of any coupling in the calculation of the CTL when using real-world data. Furthermore, any difference between coupled and uncoupled ACWRs appear to be no greater than trivial across calculation methods and sports ($g = 0$–0.08). There also may be differences in the levels of ATL–CTL correlation across internal and external TL measures and the 2 different sports examined in this case study. These findings are described in more detail below.

Similar to previous research,[3] there were significant differences between coupled and uncoupled ATL–CTL correlations. However, due to the trends in training cycles in both open- and closed-skill sports, we anticipated some level of natural correlation between ATL and CTL, irrespective of coupling. This was evident with levels of correlation between ATL and uncoupled CTL ranging from $r = .17$ to .53 as rolling averages and $r = .19$ to .35 as EWMA. The magnitude of these uncoupled correlations was much higher than those reported by Lolli et al[3,14] (randomly generated data $r = .01$, professional soccer external TL $r = -.04$) with the lowest external TL correlation reported in this study being $r = .20$. This difference may be related to a number of factors including the data sets analyzed (eg, randomly generated data vs data obtained from international-level athletes); the variables analyzed (eg, sessional ratings of perceived exertion × t and PlayerLoad/min × t vs total distance); general training practices in the different open-skill sports (eg, basketball vs football); or different coaching styles/philosophies around managing TL.

When comparing differences between coupled and uncoupled ACWRs, there were only trivial effect sizes. These trivial effect

Table 1 Summary of TL Descriptive Statistics Calculated Using Rolling Averages From Open- and Closed-Skill Sports Preceding Qualification for the 2016 Olympic Games

	Acute	Chronic	Chronic UNC	*P*	*g*	ACWR	ACWR UNC	*P*	*g*
Closed-skill internal TL	339.1 (123.9)	354.6 (91.3)	368.2 (93.8)	.05*	0.12	0.87 (0.23)	0.86 (0.28)	.28	0.06
Open-skill internal TL	619.2 (254.2)	615.6 (158.9)	622.8 (178.6)	.19	0.06	0.97 (0.45)	1.06 (0.74)	.55	0.02
Open-skill external TL	378.2 (134.3)	374.9 (84.3)	381.0 (91.7)	.04*	0.14	0.95 (0.32)	0.98 (0.43)	.94	0

Abbreviations: ACWR, acute to chronic workload ratio; *g*, Hedge *g* effect size; TL, training load; UNC, uncoupled.
*$P < .05$.

Table 2 Summary of TL Descriptive Statistics Calculated Using Exponentially Weighted Moving Averages From Open- and Closed-Skill Sports Preceding Qualification for the 2016 Olympic Games

	Acute	Chronic	Chronic UNC	*P*	*g*	ACWR	ACWR UNC	*P*	*g*
Closed-skill internal TL	311.6 (135.4)	247.9 (113.9)	278.5 (116.9)	.00**	0.24	1.13 (0.31)	1.16 (0.49)	.13	0.08
Open-skill internal TL	602.6 (255.1)	553.6 (180.6)	570.8 (195.2)	.00**	0.10	1.00 (0.33)	1.11 (0.67)	.21	0.04
Open-skill external TL	368.5 (136.1)	339.1 (101.6)	352.1 (106.2)	.00**	0.14	1.00 (0.28)	1.05 (0.45)	.97	0

Abbreviations: ACWR, acute to chronic workload ratio; *g*, Hedge *g* effect size; TL, training load; UNC, uncoupled.
*$P < .05$. **$P < .01$.

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

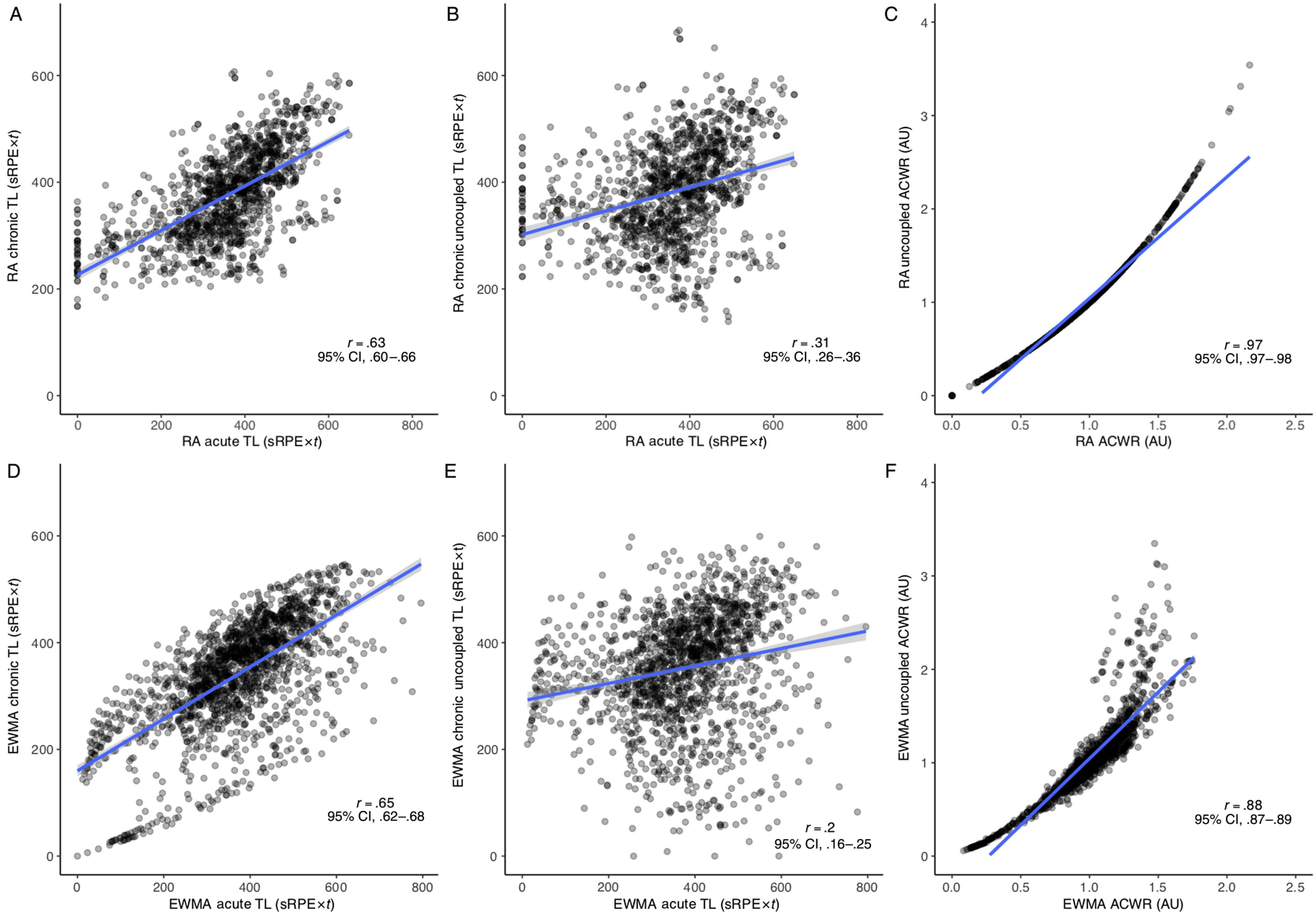

Figure 1 — The relationship between coupled and uncoupled internal TL variables using different calculation methods in a closed-skill sport preceding qualification for the 2016 Olympic Games. ACWR indicates acute to chronic workload ratio; AU, arbitrary units; CI, confidence interval; EWMA, exponentially weighted moving averages; *r*, Pearson correlation; RA, rolling averages; sRPE, session rating of perceived exertion; *t*, session duration; TL, training load.

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

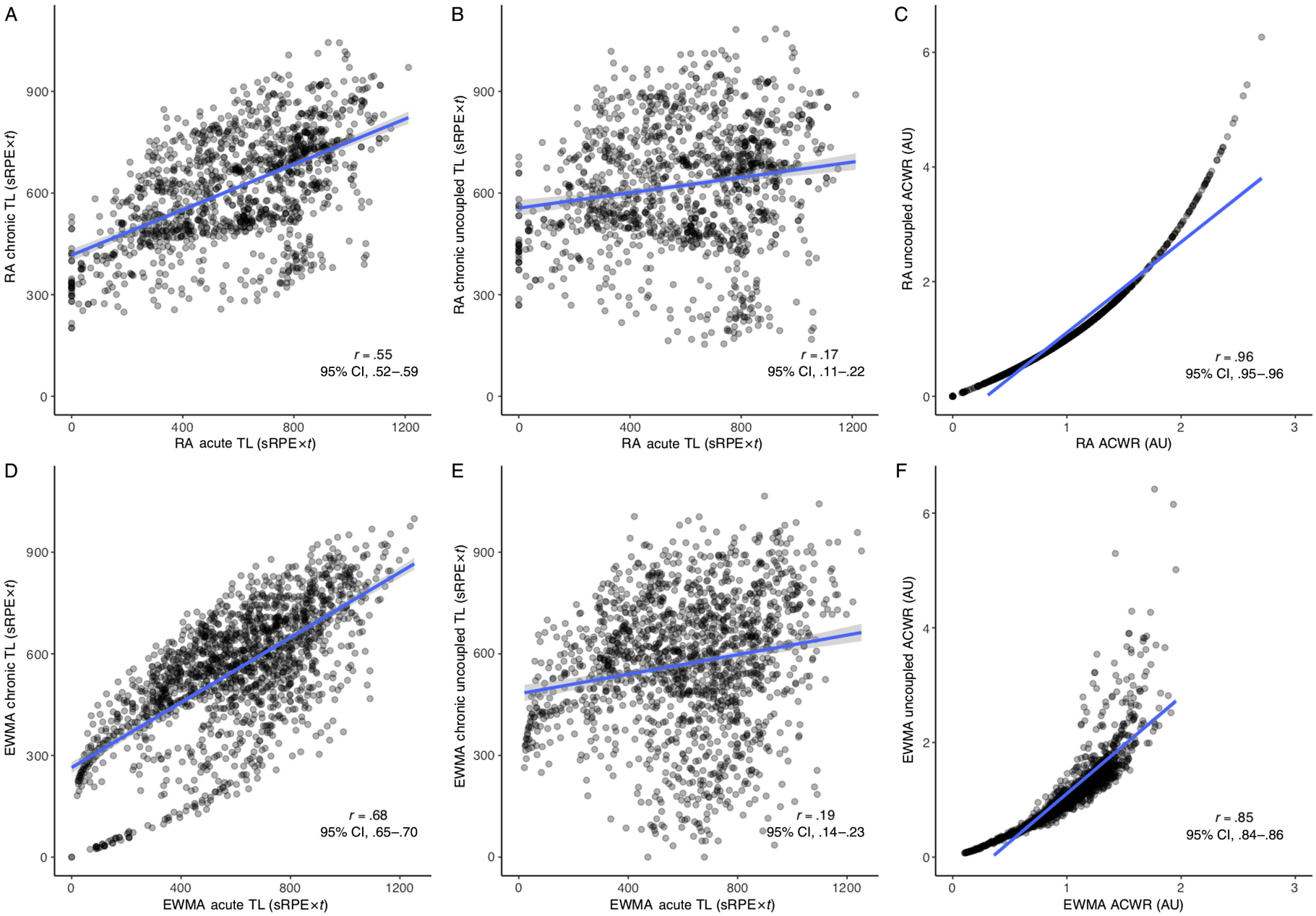

Figure 2 — The relationship between coupled and uncoupled internal TL variables using different calculation methods in an open-skill sport preceding qualification for the 2016 Olympic Games. ACWR indicates acute to chronic workload ratio; AU, arbitrary units; CI, confidence interval; EWMA, exponentially weighted moving averages; *r*, Pearson correlation; RA, rolling averages; sRPE, session rating of perceived exertion; *t*, session duration; TL, training load.

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

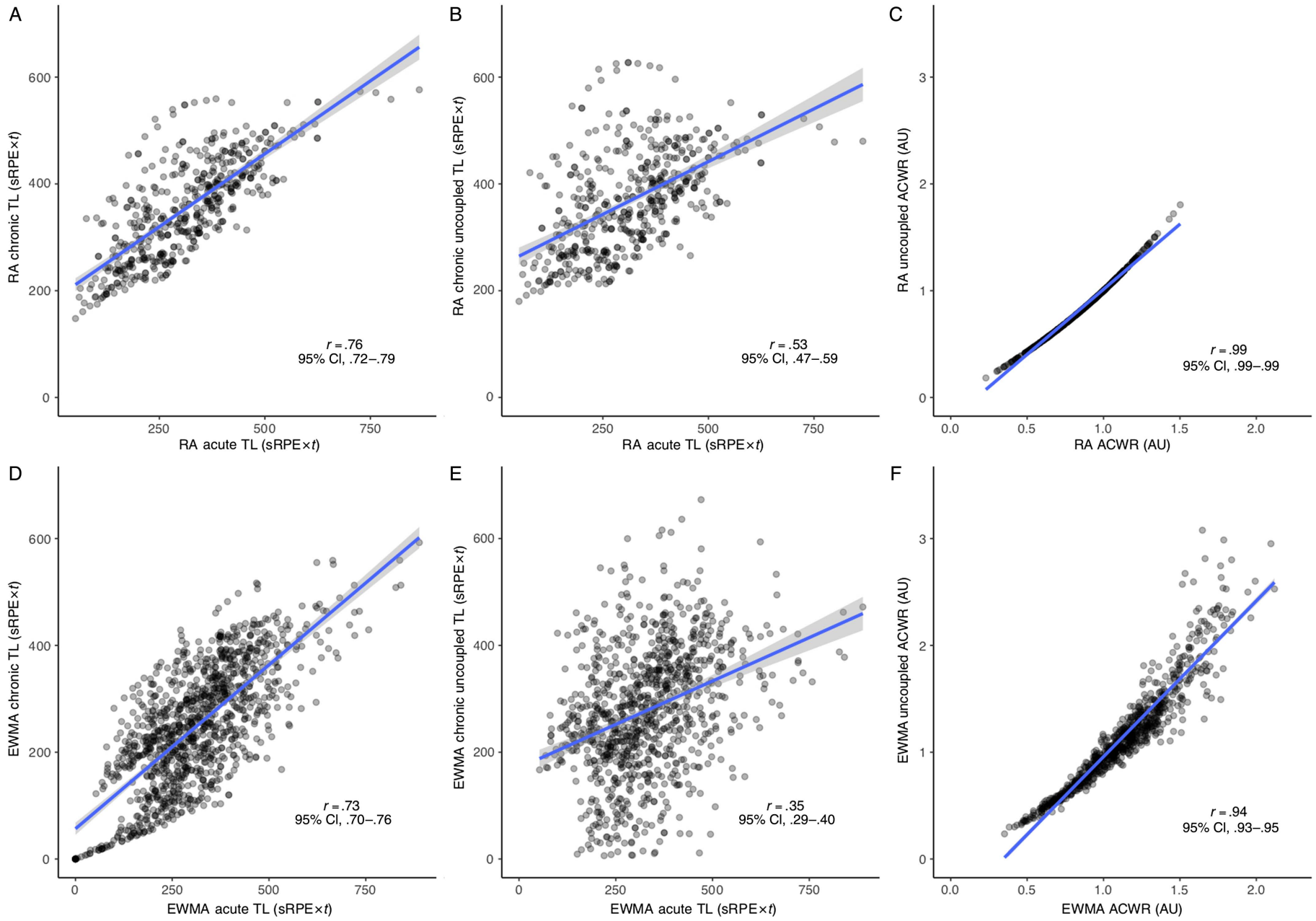

Figure 3 — The relationship between coupled and uncoupled external TL variables using different calculation methods in an open-skill sport preceding qualification for the 2016 Olympic Games. ACWR indicates acute to chronic workload ratio; AU, arbitrary units; CI, confidence interval; EWMA, exponentially weighted moving averages; PL/min, PlayerLoad per minute; *r*, Pearson correlation; RA, rolling averages; *t*, session duration; TL, training load.

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

sizes were combined with very large to nearly perfect correlations between the different forms of the ACWR in both OS and CS, depending on the calculation method. These findings show mathematical coupling may have little effect on practical measures of the ACWR in this case study. This is supported by an uncoupled ACWR demonstrating a similar association as the coupled ACWR–injury likelihood relationship in one study.[6] It is also interesting that the explained variance between rolling average ATL and CTL ranged from 31% to 58% in this study. This is considerably higher than the theoretical ~25% based on the 7:28 day ACWR time frames previously suggested in the scientific literature.[24]

In regard to the effect of internal versus external TL measures, there were lower uncoupled correlations for internal TL across both OS and CS. Internal TL is directly impacted by the athletes' response to and ability to handle the external TL.[1] If external TL stays relatively similar across training cycles, it stands to reason that internal TL may change as athlete adaptation to the external TL occurs.[1] It is also logical that the internal TL will be more variable across training cycles when considering the multitude of influences that impact the athlete, including those that are outside of the training environmental (eg, emotional, psychosocial, and environmental stressors).

Another component of this case study was the difference in TL measures between the OS and CS groups. The OS group exhibited lower correlations between coupled and uncoupled variations of CTL and the ACWR. In the authors' experience, CS sports will generally have a more routine training structure and shorter competition period than OS sports. Due to the different training and competition nature between open- and closed-skill sports, these differences between OS and CS correlations seem logical.

A potential limitation of the current examples used in this commentary is that both data sets rely on data collection periods that are relatively short for TL monitoring research. The OS group was also an international basketball squad and professional club TL data may be different due to different competitive schedules. Considering the nature of international elite sport and different coaching/training styles, it is very likely different sports and data sets within the same sport will possess different levels of ATL–CTL correlations and levels of explained variance. Furthermore, due to the data collection period length, this commentary has only assessed the relationship between coupled and uncoupled TL measures; not the TL–injury etiological relationship itself. More research is warranted comparing coupled and uncoupled internal/external TL measures' relationship with injury across different sports. Practitioners should also interpret this commentary's findings as a case study.

Practical Applications

In practice, both coupled and uncoupled ACWR calculations give practitioners a single measure on whether recent athlete workloads are increasing or decreasing relative to prior workloads.[24] Mathematical coupling only becomes spurious if it is not initially accounted for and understood.[4,13] As such, practitioners should understand how mathematical coupling affects both ACWR calculation methods (for further information, see Windt and Gabbett[24]). If practitioners decide to use the ACWR for athlete monitoring purposes despite its potential limitations, it is suggested the ACWR should not be used solely in isolation for athlete monitoring purposes. Instead, it should be incorporated into a multifactorial model,[24] preferably on a daily basis. Practitioners should also be aware of the numerous factors (eg, strength, previous injury, aerobic fitness, genetics) that may impact the ACWR's relationship with injury.[8] Furthermore, there should be the consideration that individual athletes may respond differently to any given combination of ATL, CTL, or ACWR and individualized monitoring is recommended.[25]

Conclusions

Any real-world measure in general will contain a combination of contributing measures that may all have some degree of correlation. In this case, athletes often repeat similar training cycles within defined phases (eg, general preparation, specific preparation and competition), and it appears that varying levels of correlation may exist naturally between ATL and CTL across open- and closed-skill sports. Judging from this case study, mathematical coupling may have little effect on athletes' ACWR measures. It may be reasonable for practitioners wanting to apply the ACWR to be able to choose which version suits their purposes best. However, practitioners are cautioned that conceptually the uncoupled ACWR possesses a lag period that disregards the most recent CTL an athlete has completed. This lag period is more pronounced if practitioners choose to use EWMA as their TL smoothing method. Finally, using either version of the ACWR for injury prediction requires more research, and we welcome future investigations on this topic. As further evidence from these investigations emerges, we also encourage practitioners to consider the context of the research, especially in regard to the nature of the sport, athletes, and variables examined.

Acknowledgments

J.O.C.C. drafted the initial concept, and all authors contributed to drafting, critical revision, and approval of the final manuscript. There was no additional financial or material support.

References

1. Bourdon PC, Cardinale M, Murray A, et al. Monitoring athlete training loads: consensus statement. *Int J Sports Physiol Perform*. 2017;12(suppl 2):161–170. PubMed ID: 28463642 doi:10.1123/IJSPP.2017-0208
2. Soligard T, Schwellnus M, Alonso JM, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *Br J Sports Med*. 2016;50(17):1030–1041. PubMed ID: 27535989 doi:10.1136/bjsports-2016-096581
3. Lolli L, Batterham AM, Hawkins R, et al. Mathematical coupling causes spurious correlation within the conventional acute-to-chronic workload ratio calculations. *Br J Sports Med*. 2019;53(15):921–922. PubMed ID: 29101104 doi:10.1136/bjsports-2017-098110
4. Archie JP. Mathematic coupling of data: a common source of error. *Ann Surg*. 1981;193(3):296–303. PubMed ID: 7212790 doi:10.1097/00000658-198103000-00008
5. Raya-Gonzalez J, Nakamura FY, Castillo D, Yanci J, Fanchini M. Determining the relationship between internal load markers and non-contact injuries in young elite soccer players. *Int J Sports Physiol Perform*. 2019;14(4):421–425. PubMed ID: 30204512 doi:10.1123/ijspp.2018-0466
6. Weiss KJ, Allen SV, McGuigan MR, Whatman CS. The relationship between training load and injury in men's professional basketball. *Int J Sports Physiol Perform*. 2017;12(9):1238–1242. PubMed ID: 28253031 doi:10.1123/ijspp.2016-0726

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC

7. Bowen L, Gross AS, Gimpel M, Bruce-Low S, Li FX. Spikes in acute:chronic workload ratio (ACWR) associated with a 5–7 times greater injury rate in English Premier League football players: a comprehensive 3-year study [published online ahead of print February 21, 2019]. *Br J Sports Med*. doi:10.1136/bjsports-2018-099422
8. Coyne JOC, Gregory Haff G, Coutts AJ, Newton RU, Nimphius S. The current state of subjective training load monitoring—a practical perspective and call to action. *Sports Med Open*. 2018;4(1):58. PubMed ID: 30570718 doi:10.1186/s40798-018-0172-x
9. Windt J, Ardern CL, Gabbett TJ, et al. Getting the most out of intensive longitudinal data: a methodological review of workload–injury studies. *BMJ Open*. 2018;8(10):e022626. PubMed ID: 30282683 doi:10.1136/bmjopen-2018-022626
10. Banister EW, Calvert TW, Savage MV, Bach TM. A systems model of training for athletic performance. *Aust J Sports Med*. 1975;7:57–61.
11. Williams S, West S, Cross MJ, Stokes KA. Better way to determine the acute:chronic workload ratio? *Br J Sports Med*. 2016; 51(3):209–210. PubMed ID: 27650255 doi:10.1136/bjsports-2016-096589
12. Murray NB, Gabbett TJ, Townshend AD, Blanch P. Calculating acute:chronic workload ratios using exponentially weighted moving averages provides a more sensitive indicator of injury likelihood than rolling averages. *Br J Sports Med*. 2016;51(9):749–754. PubMed ID: 28003238 doi:10.1136/bjsports-2016-097152
13. Tu YK, Gilthorpe MS. Revisiting the relation between change and initial value: a review and evaluation. *Stat Med*. 2007;26(2):443–457. PubMed ID: 16526009 doi:10.1002/sim.2538
14. Lolli L, Batterham AM, Hawkins R, et al. The acute-to-chronic workload ratio: an inaccurate scaling index for an unnecessary normalisation process? [published online ahead of print June 13, 2018]. *Br J Sports Med*. doi:10.1136/bjsports-2017-098884
15. Impellizzeri FM, Rampinini E, Marcora SM. Physiological assessment of aerobic training in soccer. *J Sports Sci*. 2005;23(6):583–592. PubMed ID: 16195007 doi:10.1080/02640410400021278
16. Foster C, Florhaug JA, Franklin J, et al. A new approach to monitoring exercise training. *J Strength Cond Res*. 2001;15(1):109–115. PubMed ID: 11708692
17. Wik EH, Luteberget LS, Spencer M. Activity profiles in international women's team handball using PlayerLoad. *Int J Sports Physiol Perform*. 2017:12(7):934–942. doi:10.1123/ijspp.2015-0732
18. Barrett S, Midgley A, Lovell R. PlayerLoad: reliability, convergent validity, and influence of unit position during treadmill running. *Int J Sports Physiol Perform*. 2014;9(6):945–952. PubMed ID: 24622625 doi:10.1123/ijspp.2013-0418
19. Fuller CW, Ekstrand J, Junge A, et al. Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries. *Br J Sports Med*. 2006;40(3):193–201. PubMed ID: 16505073 doi:10.1136/bjsm.2005.025270
20. Fuller CW, Molloy MG, Bagate C, et al. Consensus statement on injury definitions and data collection procedures for studies of injuries in rugby union. *Br J Sports Med*. 2007;41(5):328–331. PubMed ID: 17452684 doi:10.1136/bjsm.2006.033282
21. Raysmith BP, Drew MK. Performance success or failure is influenced by weeks lost to injury and illness in elite Australian Track and Field athletes: a 5-year prospective study. *J Sci Med Sport*. 2016;19(10):778–783. PubMed ID: 26839047 doi:10.1016/j.jsams.2015.12.515
22. Rosenthal R. Parametric measures of effect size. In: Cooper H, Hedges LV, eds. *The Handbook of Research Synthesis*. New York, NY: Sage; 1994:231–244.
23. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41(1):3–13. PubMed ID: 19092709 doi:10.1249/MSS.0b013e31818cb278
24. Windt J, Gabbett TJ. Is it all for naught? What does mathematical coupling mean for acute:chronic workload ratios? *Br J Sports Med*. 2019;53(16):988–990. doi:10.1136/bjsports-2017-098925
25. Ward P, Coutts AJ, Pruna R, McCall A. Putting the "I" back in team. *Int J Sports Physiol Perform*. 2018;13(8):1107–1111. PubMed ID: 29952669 doi:10.1123/ijspp.2018-0154

Authenticated t_lambrinides@yahoo.com | Downloaded 12/12/19 04:03 PM UTC